Программчлалын хэл

Лабораторын ажил 12

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1. Хоёр тооны үйлдэл хийх модуль

```
LAB_12
  ex1
    a
    calculator.c
    calculator.h
    main.c
```

ex1 > main.c > main()

```c
#include <stdio.h>
#include "calculator.h"

int main(){
    int a, b;
    printf("2 too oruulna uu: ");
    scanf("%d %d", &a, &b);
    printf("Niilber: %d\n", add(a, b));
    printf("Yalgavar: %d\n", subtract(a, b));
    printf("Urjver: %d\n", multiply(a, b));
    printf("Noogdvor: %.2f\n", divide(a, b));
    return 0;
}
```

ex1 > calculator.c > add(int, int)

```c
#include <stdio.h>
#include "calculator.h"
int add(int a, int b){
    return a+b;
}
int subtract(int a, int b){
    return a-b;
}
int multiply(int a, int b){
    return a*b;
}
double divide(int a, int b){
    if(b==0){
        printf("Too 0-d huvaagdahgui.");
        return 0;
    } else {
        return (double)a/b;
    }
}
```

ex1 > calculator.h > ...

```c
#ifndef CALCULATOR_H
#define CALCULATOR_H

int add (int a, int b);
int subtract(int a, int b);
int multiply(int a, int b);
double divide(int a, int b);

#endif
```

Гаралт:

```
macbook@macbooks-MacBook-Air ex1 % clang main.c calculator.c -o a
macbook@macbooks-MacBook-Air ex1 % ./a
2 too oruulna uu: 6 7
Niilber: 13
Yalgavar: -1
Urjver: 42
Noogdvor: 0.86
macbook@macbooks-MacBook-Air ex1 %
```

2. Массиваар ажиллах модуль

```
ex2
  a
  array_utils.c
  array_utils.h
  main.c
```

array_utils.c | array_utils.h | main.c

ex2 > array_utils.h > ...

```c
#ifndef ARRAY_UTILS_H
#define ARRAY_UTILS_H
int find_sum(int arr[], int n);
int find_max(int arr[], int n);
int find_min(int arr[], int n);
double find_average(int arr[], int n);
void print_array(int arr[], int n);
#endif
```

ex2 > C array_utils.c > print_array(int [], int)

```c
#include <stdio.h>
#include "array_utils.h"
int find_sum(int arr[], int n){
    int sum = 0;
    for (int i = 0; i < n; i++){
        sum=sum+arr[i];
    }
    return sum;
}
int find_max(int arr[], int n){
    int max = arr[0];
    for (int i = 0; i < n; i++){
        if (arr[0]<arr[i]){
            max = arr[i];
        }
    }
    return max;
}
int find_min(int arr[], int n){
    int min = arr[0];
    for (int i = 0; i < n; i++){
        if (arr[0]>arr[i]){
            min=arr[i];
        }
    }
    return min;
}
double find_average(int arr[], int n){
    double average=find_sum(arr, n)/n;
    return average;
}
void print_array(int arr[], int n){
    printf("array");
    for (int i = 0; i < n; i++){
        printf("%d", arr[i]);
    }
}
```

ex2 > C main.c > main()

```c
#include <stdio.h>
#include "array_utils.h"
int main(){
    int n;
    printf("Heden too oruulah ve: ");
    scanf("%d", &n);
    int a[n];
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }
    print_array(a, n);
    printf("Niilber: %d\n", find_sum(a, n));
    printf("Hamgiin baga: %d\n", find_min(a, n));
    printf("Hamgiin ih: %d\n", find_max(a, n));
    printf("Dundaj: %f", find_average(a, n));
    return 0;
}
```

Гаралт:

```
macbook@macbooks-MacBook-Air ex2 % clang array_utils.c main.c -o a
macbook@macbooks-MacBook-Air ex2 % ./a
Heden too oruulah ve: 5
2 6 4 9 1
array26491Niilber: 22
Hamgiin baga: 1
Hamgiin ih: 9
Dundaj: 4.000000%
macbook@macbooks-MacBook-Air ex2 %
```

3. Тоо шинжлэх модуль

- ex3
  - a
  - main.c
  - number_utils.c
  - number_utils.h

ex3 > C number_utils.h > ...

```c
#ifndef NUMBERS_UTILS_H
#define NUMBERS_UTILS_H
int is_even(int n);
int is_prime(int n);
int factorial(int n);
int digit_sum(int n);
#endif
```

ex3 > C number_utils.c > is_even(int)

```c
#include <stdio.h>
#include "number_utils.h"
int is_even(int n){
    if (n%2==0)
    {
        return 1;
    }else {
        return 0;
    }
    return 0;
}
int is_prime(int n) {
    if (n <= 1) return 0;
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            return 0;
        }
    }
    return 1;
}
int factorial(int n) {
    if (n < 0) return -1;

    int result = 1;
    for (int i = 1; i <= n; i++) {
        result *= i;
    }
    return result;
}

int digit_sum(int n) {
    if (n < 0) n = -n;
    int sum = 0;
    while (n > 0) {
        sum += n % 10;
        n /= 10;
    }
    return sum;
}
```

ex3 > C main.c > main()

```c
#include <stdio.h>
#include "number_utils.h"
int main(){
    int n;
    printf("Too oruulna uu: ");
    scanf("%d", &n);
    printf("Tegsh too eseh: %d\n", is_even(n));
    printf("Anhnii too eseh: %d\n", is_prime(n));
    printf("Factorial: %d\n",factorial(n));
    printf("Digit sum: %d\n", digit_sum(n));
    return 0;
}
```

Гаралт:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
macbook@macbooks-MacBook-Air ex3 % clang main.c number_utils.c -o a
macbook@macbooks-MacBook-Air ex3 % ./a
Too oruulna uu: 5
Tegsh too eseh: 0
Anhnii too eseh: 1
Factorial: 120
Digit sum: 5
macbook@macbooks-MacBook-Air ex3 %
```

4. String боловсруулах модуль

- ex4
  - a
  - main.c
  - string_utils.c
  - string_utils.h

C string_utils.h

ex4 > C string_utils.h > ...

```c
#ifndef STRING_UTILS_H
#define STRING_UTILS_H
int str_length(char s[]);
int count_vowels(char s[]);
void to_uppercase(char s[]);
#endif
```

C main.c

ex4 > C main.c > main()

```c
#include <stdio.h>
#include "string_utils.h"
int main(){
    char s[101];
    printf("Ug oruulna uu: ");
    fgets(s, sizeof(s), stdin);
    for (int i = 0; s[i] !='\0'; i++)
    {
        if (s[i]=='\n')
        {
            s[i]='\0';
            break;
        }
    }
    printf("Ugiin urt: %d\n", str_length(s));
    printf("Egshigiin too: %d\n", count_vowels(s));
    to_uppercase(s);
    printf("Tom useg: %s", s);
    return 0;
}
```

```c
#include "string_utils.h"
#include <ctype.h>
int str_length(char s[]) {
    int length = 0;
    while (s[length] != '\0') {
        length++;
    }
    return length;
}

int count_vowels(char s[]) {
    int count = 0;
    int i = 0;
    while (s[i] != '\0') {
        char c = s[i];
        if (c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u' ||
            c == 'A' || c == 'E' || c == 'I' || c == 'O' || c == 'U') {
            count++;
        }
        i++;
    }
    return count;
}

void to_uppercase(char s[]) {
    int i = 0;
    while (s[i] != '\0') {
        if (s[i] >= 'a' && s[i] <= 'z') {
            s[i] = s[i] - 32;
        }
        i++;
    }
}
```

Гаралт:

```
macbook@macbooks-MacBook-Air ex4 % clang main.c string_utils.c -o a
macbook@macbooks-MacBook-Air ex4 % ./a
Ug oruulna uu: Hi C language
Ugiin urt: 13
Egshigiin too: 5
Tom useg: HI C LANGUAGE%
macbook@macbooks-MacBook-Air ex4 %
```

5. Геометрийн модуль

```
ex5
  a
  geometry.c
  geometry.h
  main.c
```

```c
#ifndef GEOMETRY_H
#define GEOMETRY_H
#define pi 3.1415926
double area_rectangle(double a, double b);
double perimeter_rectangle(double a, double b);
double area_circle(double r);
double circumference(double r);
#endif
```

```c
#include "geometry.h"
double area_rectangle(double a, double b){
    return a*b;
}
double perimeter_rectangle(double a, double b){
    return 2*(a+b);
}
double area_circle(double r){
    return pi*r*r;
}
double circumference(double r){
    return 2*pi*r;
}

```

```c
#include <stdio.h>
#include "geometry.h"
int main(){
    int a, b, r;
    printf("tegsh untsugtiin taluudiig oruulna uu: ");
    scanf ("%d %d", &a, &b);
    printf("toirgiin radius oruulna uu: ");
    scanf("%d", &r);
    printf("tegsh untsugtiin talbai: %.2f\n", area_rectangle(a, b));
    printf("tegsh untsugtiin perimeter: %.2f\n", perimeter_rectangle(a, b));
    printf("toirgiin talbai: %.2f\n", area_circle(r));
    printf("toirgiin urt: %.2f", circumference(r));
}
```

Гаралт:

```
macbook@macbooks-MacBook-Air ex5 % clang main.c geometry.c -o a
macbook@macbooks-MacBook-Air ex5 % ./a
tegsh untsugtiin taluudiig oruulna uu: 4 7
toirgiin radius oruulna uu: 5
tegsh untsugtiin talbai: 28.00
tegsh untsugtiin perimeter: 22.00
toirgiin talbai: 78.54
toirgiin urt: 31.42%
macbook@macbooks-MacBook-Air ex5 %
```

6. Менютэй модульчлагдсан программ

```
ex6
  a
  array_utils.c
  array_utils.h
  calculator.c
  calculator.h
  main.c
  number_utils.c
  number_utils.h
```

```c
#include <stdio.h>
#include "calculator.h"
#include "array_utils.h"
#include "number_utils.h"

int main() {
    int choice;
    do {
        printf("\n==== MENU ====\n");
        printf("1. Calculator\n");
        printf("2. Array\n");
        printf("3. Number analyze\n");
        printf("4. Exit\n");
        printf("Songolt: ");
        scanf("%d", &choice);
        if (choice == 1) {
        double a, b;
        int op;
        printf("a, b: ");
        scanf("%lf %lf", &a, &b);
        while(getchar() != '\n');
        printf("1:+ 2:- 3:* 4:/ : ");
        scanf("%d", &op);
        while(getchar() != '\n');
        if(op == 1) printf("Result: %.2lf\n", add(a,b));
        else if(op == 2) printf("Result: %.2lf\n", sub(a,b));
        else if(op == 3) printf("Result: %.2lf\n", mul(a,b));
        else if(op == 4) printf("Result: %.2lf\n", div(a,b));
        else printf("Invalid!\n");
    }
```

```c
        else if(op == 2) printf("Result: %.2lf\n", sub(a,b));
        else if(op == 3) printf("Result: %.2lf\n", mul(a,b));
        else if(op == 4) printf("Result: %.2lf\n", div(a,b));
        else printf("Invalid!\n");
    }
        else if (choice == 2) {
            int n;
            printf("Hemjee: ");
            scanf("%d", &n);
            int arr[n];
            for(int i = 0; i < n; i++) {
                printf("arr[%d]: ", i);
                scanf("%d", &arr[i]);
            }
            printf("Sum = %d\n", find_sum(arr, n));
            printf("Max = %d\n", find_max(arr, n));
        }
        else if (choice == 3) {
            int n;
            printf("Too: ");
            scanf("%d", &n);

            printf("Prime: %s\n", is_prime(n) ? "Yes" : "No");
            printf("Factorial: %d\n", factorial(n));
        }
    } while (choice != 4);

    return 0;
}
```

Гаралт:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

macbook@macbooks-MacBook-Air ex6 % clang main.c calculator.c number_utils.c array_utils.c -o a
macbook@macbooks-MacBook-Air ex6 % ./a

  ==== MENU ====
  1. Calculator
  2. Array
  3. Number analyze
  4. Exit
  Songolt: 1
  a, b: 4 5
  1:+ 2:- 3:* 4:/ : 3
  Result: 20.00

  ==== MENU ====
  1. Calculator
  2. Array
  3. Number analyze
  4. Exit
  Songolt: 4
macbook@macbooks-MacBook-Air ex6 %
```

7. Student модуль

```
ex7
  a
  main.c
  student.c
  student.h
```

student.c M | student.h M

ex7 > student.h > ...

```c
#ifndef STUDENT_H
#define STUDENT_H
typedef struct {
    char name[50];
    int age;
    float gpa;
} Student;
void input_student(Student *s);
void print_student(Student s);
int is_honor(Student s);
#endif
```

student.c M | student.h M

ex7 > student.c > is_honor(Student)

```c
#include <stdio.h>
#include <string.h>
#include "student.h"
void input_student(Student *s) {
    printf("Name: ");
    fgets(s->name, sizeof(s->name), stdin);
    s->name[strcspn(s->name, "\n")] = 0;
    printf("Age: ");
    scanf("%d", &s->age);
    printf("GPA: ");
    scanf("%f", &s->gpa);
    while(getchar() != '\n');
}
void print_student(Student s) {
    printf("\n--- Student Info ---\n");
    printf("Name: %s\n", s.name);
    printf("Age: %d\n", s.age);
    printf("GPA: %.2f\n", s.gpa);
}
int is_honor(Student s) {
    return s.gpa >= 3.5;
}
```

student.c M | main.c M | student.h M

ex7 > main.c > main()

```c
#include <stdio.h>
#include "student.h"

int main() {
    Student s;
    input_student(&s);
    print_student(s);
    if (is_honor(s))
        printf("Honor student\n");
    else
        printf("Not honor student\n");

    return 0;
}
```

Гаралт:

```
macbook@macbooks-MacBook-Air ex7 % clang main.c student.c -o a
macbook@macbooks-MacBook-Air ex7 % ./a
Name: davaadulam
Age: 18
GPA: 3.2

--- Student Info ---
Name: davaadulam
Age: 18
GPA: 3.20
Not honor student
macbook@macbooks-MacBook-Air ex7 %
```

8.Файлаас унших модуль

```
ex8
  a.out
  file_utils.c
  file_utils.h
  main.c
  numbers.txt
```

file_utils.h

ex8 > file_utils.h > ...

```c
#ifndef FILE_UTILS_H
#define FILE_UTILS_H

void write_numbers(const char *filename, int arr[], int n);
void read_numbers(const char *filename);

#endif
```

student.c | main.c | file_utils.c | student.h

ex8 > file_utils.c > write_numbers(const char *, int [], int)

```c
#include <stdio.h>
#include "file_utils.h"
void write_numbers(const char *filename, int arr[], int n) {
    FILE *fp = fopen(filename, "w");
    if (fp == NULL) {
        printf("File neehad aldaa!\n");
        return;
    }
    for (int i = 0; i < n; i++) {
        fprintf(fp, "%d ", arr[i]);
    }
    fclose(fp);
}
void read_numbers(const char *filename) {
    FILE *fp = fopen(filename, "r");
    if (fp == NULL) {
        printf("File oldsongui!\n");
        return;
    }
    int x;
    printf("File dotorh toonuud:\n");
    while (fscanf(fp, "%d", &x) != EOF) {
        printf("%d ", x);
    }
    printf("\n");
    fclose(fp);
}
```

main.c M

ex8 > main.c > main()

```c
#include <stdio.h>
#include "file_utils.h"

int main() {
    int arr[] = {10, 20, 30, 40, 50};
    int n = 5;

    char filename[] = "numbers.txt";
    write_numbers(filename, arr, n);
    read_numbers(filename);

    return 0;
}
```

Гаралт:

```
macbook@macbooks-MacBook-Air ex8 % clang main.c file_utils.c -o a
macbook@macbooks-MacBook-Air ex8 % ./a
File dotorh toonuud:
10 20 30 40 50
macbook@macbooks-MacBook-Air ex8 %
```

9. Номын сангийн жижиг систем

```
ex9
  a
  book.c
  book.h
  books.txt
  file_utils.c
  file_utils.h
  main.c
```

main.c M | file_utils.h

ex9 > file_utils.h > ...

```c
#ifndef FILE_UTILS_H
#define FILE_UTILS_H

#include "book.h"

void save_books(const char *filename, Book arr[], int n);
int load_books(const char *filename, Book arr[]);

#endif
```

main.c M | book.h

ex9 > book.h > ...

```c
#ifndef BOOK_H
#define BOOK_H

typedef struct {
    int id;
    char title[100];
    char author[50];
    int year;
} Book;

void input_book(Book *b);
void print_book(Book b);
void list_books(Book arr[], int n);
int find_book(Book arr[], int n, int id);

#endif
```

book.c

ex9 > book.c > print_book(Book)

```c
#include <stdio.h>
#include <string.h>
#include "book.h"

void input_book(Book *b) {
    printf("ID: ");
    scanf("%d", &b->id);
    while(getchar() != '\n');

    printf("Title: ");
    fgets(b->title, sizeof(b->title), stdin);
    b->title[strcspn(b->title, "\n")] = 0;

    printf("Author: ");
    fgets(b->author, sizeof(b->author), stdin);
    b->author[strcspn(b->author, "\n")] = 0;

    printf("Year: ");
    scanf("%d", &b->year);
    while(getchar() != '\n');
}
```

```c

void print_book(Book b) {
    printf("\nID: %d\nTitle: %s\nAuthor: %s\nYear: %d\n",
            b.id, b.title, b.author, b.year);
}

void list_books(Book arr[], int n) {
    for(int i = 0; i < n; i++) {
        print_book(arr[i]);
    }
}

int find_book(Book arr[], int n, int id) {
    for(int i = 0; i < n; i++) {
        if(arr[i].id == id)
            return i;
    }
    return -1;
}
```

file_utils.c

ex9 > file_utils.c > load_books(const char *, Book [])

```c
#include <stdio.h>
#include "file_utils.h"

void save_books(const char *filename, Book arr[], int n) {
    FILE *fp = fopen(filename, "w");
    if (fp == NULL) {
        printf("File aldaa!\n");
        return;
    }

    for(int i = 0; i < n; i++) {
        fprintf(fp, "%d|%s|%s|%d\n",
                arr[i].id,
                arr[i].title,
                arr[i].author,
                arr[i].year);
    }

    fclose(fp);
}
```

```c

int load_books(const char *filename, Book arr[]) {
    FILE *fp = fopen(filename, "r");
    if (fp == NULL) {
        printf("File oldsongui!\n");
        return 0;
    }

    int n = 0;

    while (fscanf(fp, "%d|%99[^|]|%49[^|]|%d\n",
            &arr[n].id,
            arr[n].title,
            arr[n].author,
            &arr[n].year) == 4) {
        n++;
    }

    fclose(fp);
    return n;
}
```

main.c

ex9 > main.c > main()

```c
#include <stdio.h>
#include "book.h"
#include "file_utils.h"

#define MAX 100

int main() {
    Book books[MAX];
    int count = 0;
    int choice;
    char filename[] = "books.txt";

    do {
        printf("\n==== LIBRARY MENU ====\n");
        printf("1. Add book\n");
        printf("2. List books\n");
        printf("3. Find by ID\n");
        printf("4. Save to file\n");
        printf("5. Load from file\n");
        printf("6. Exit\n");
        printf("Choice: ");
        scanf("%d", &choice);
        while(getchar() != '\n');

        if (choice == 1) {
            input_book(&books[count]);
            count++;
        }

        else if (choice == 2) {
            list_books(books, count);
        }
```

```c

        else if (choice == 3) {
            int id;
            printf("Enter ID: ");
            scanf("%d", &id);

            int idx = find_book(books, count, id);
            if (idx != -1)
                print_book(books[idx]);
            else
                printf("Not found!\n");
        }

        else if (choice == 4) {
            save_books(filename, books, count);
            printf("Saved!\n");
        }

        else if (choice == 5) {
            count = load_books(filename, books);
            printf("Loaded! (%d books)\n", count);
        }

    } while (choice != 6);

    return 0;
}
```

Гаралт:

```
macbook@macbooks-MacBook-Air ex9 % clang main.c book.c file_utils.c -o a
macbook@macbooks-MacBook-Air ex9 % ./a

==== LIBRARY MENU ====
1. Add book
2. List books
3. Find by ID
4. Save to file
5. Load from file
6. Exit
Choice: 1
ID: 1
Title: Collective Edge
Author: Colin M. Fisher
Year: 2025

==== LIBRARY MENU ====
1. Add book
2. List books
3. Find by ID
4. Save to file
5. Load from file
6. Exit
Choice: 2

ID: 1
Title: Collective Edge
Author: Colin M. Fisher
Year: 2025
```

```
==== LIBRARY MENU ====
1. Add book
2. List books
3. Find by ID
4. Save to file
5. Load from file
6. Exit
Choice: 4
Saved!

==== LIBRARY MENU ====
1. Add book
2. List books
3. Find by ID
4. Save to file
5. Load from file
6. Exit
Choice: 5
Loaded! (1 books)

==== LIBRARY MENU ====
1. Add book
2. List books
3. Find by ID
4. Save to file
5. Load from file
6. Exit
Choice: 6
```